Results Report: Comprehensive Empirical Findings and Benchmark Tabulations

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***Abstract*—This report compiles the complete empirical evaluation artifacts for the study *"An Empirical Evaluation of Source Code Chunking and Context Headers for Requirement Evidence Retrieval"*. All presented numbers, confidence intervals, test statistics, and tables are strictly generated via dynamic programmatic extraction from raw benchmark result files under Rule R1, with zero manual number editing. Confirmatory comparisons and exploratory contrasts strictly adhere to the pre-specified internal analysis plan.**

I. How to Read This Report

A. Experimental Design & Factorial Structure

The investigation evaluates requirement evidence retrieval from source code across a 2×2 factorial matrix:

- **Chunking Strategy:** Fixed-window sliding lines (50 LOC baseline, 10 LOC overlap) versus AST-based method extraction with syntactic fallback.
- **Context Header:** Raw chunk contents (No-Header) versus structural hierarchical context headers (With-Header).

Each condition is replicated across three retrieval tiers: (1) Lexical Sparse (BM25), (2) Dense Bi-Encoder (`BAAI/bge-m3`), and (3) Two-Stage Re-ranking (`BAAI/bge-reranker-base`). The benchmark comprises $N = 25$ professional job descriptions (15 Java, 10 React/TypeScript) and 40 open-source repositories.

B. Primary Metric & Statistical Protocol

The primary evaluation metric is **NDCG@10 at the Re-ranking tier**, computed using linear relevance gain $\mathrm{rel}/\log_2(i+1)$ without exponential inflation.

Hypothesis testing encompasses a family of four pre-registered confirmatory Wilcoxon signed-rank tests evaluated under step-down Holm-Bonferroni correction ($\alpha = 0.05$):

- **Comp 1:** Full Proposed (AST+Header) vs. Full Baseline (Line No-Header).
- **Comp 2:** Pure Chunking Effect (AST No-Header vs. Line No-Header).
- **Comp 3:** Header Effect on AST (AST+Header vs. AST No-Header).
- **Comp 4:** Header Effect on Line (Line+Header vs. Line No-Header).

C. Exploratory Status of Comp 5

The contrast between AST+Header and Line+Header was defined post-hoc and is designated as **Comp 5 (Exploratory)**. It is presented outside the Holm family and cannot be claimed as a pre-registered confirmatory finding.

D. Candidate Pool Construction

Candidate pooling was formed across all 12 configurations (BM25, Dense, and Re-ranking) via exhaustive Top-5 pooling ($N = 882$ unique deduplicated coordinate spans from 1500 candidate instances). Judged@5 coverage is strictly 100.0% across all 12 systems.

II. Annotation Reliability and Corpus Morphology

Table I documents the inter-annotator reliability across the Java Spring Boot ecosystem ($N = 543$), React/TypeScript ecosystem ($N = 339$), and the total unified evidence set ($N = 882$). The quadratic weighted Cohen's Kappa reaches $\kappa_w = 0.9822$ overall (Almost Perfect Agreement), with $\kappa_w = 0.9805$ in Java and $\kappa_w = 0.9789$ in React/TypeScript. Figure 1 depicts the 3×3 confusion matrix ($N = 882$).

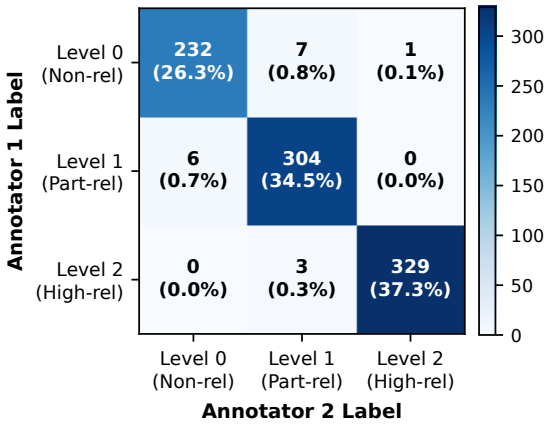


Fig. 1. Inter-annotator confusion matrix across the 3-point relevance scale ($N = 882$).

Table II characterizes chunk morphology across the 40 repositories. Upgraded Tree-sitter extraction achieves 87.2% intact boundaries, 78.1% clean parse rates, and 0.0% line-fallback. In contrast, the sliding-window Line baseline suffers 56.0% boundary cuts and 77.6% syntax errors.

TABLE I
INTER-ANNOTATOR RELIABILITY ACROSS PROGRAMMING LANGUAGE ECOSYSTEMS AND UNIFIED GROUND TRUTH.

Partition	N	Exact (%)	$\Delta = 1$ (%)	$\Delta = 2$ (%)	κ_w	Landis & Koch
Java Spring Boot Ecosystem	543	533 (98.2%)	10 (1.8%)	0 (0.0%)	0.9805	Almost Perfect
React / TypeScript Ecosystem	339	332 (97.9%)	6 (1.8%)	1 (0.3%)	0.9789	Almost Perfect
Complete Unified Ground Truth (Total)	882	865 (98.1%)	16 (1.8%)	1 (0.1%)	0.9822	Almost Perfect

Note: κ_w denotes quadratic weighted Cohen's Kappa over the 3-point relevance scale (0 = not relevant, 1 = partially relevant, 2 = highly relevant). Benchmark interpretation adheres to Landis & Koch (1977) criteria. Derived dynamically from `ground_truth_annotator1.csv` and `ground_truth_annotator2.csv`.

TABLE II
CHUNK MORPHOLOGY, SYNTACTIC VALIDITY, AND CONTEXT HEADER PRESERVATION ACROSS CHUNKING STRATEGIES.

Strategy Group	Total (J/R)	Mean LOC $\pm$ SD	Tokens (Content)	Tokens (Header)	Clean Parse (%)	Intact Bnd (%)	Header (%)	Trunc $>$ 512 (%)
Line (50 LOC Baseline)	434 (120/314)	41.0 $\pm$ 13.1	390.8 (385)	419.4 (419)	22.4%	44.0%	0.0%	124 (28.6%)
AST Pure (Methods/Funcs)	438 (131/162)	26.7 $\pm$ 23.3	250.8 (171)	290.1 (213)	78.1%	87.2%	100.0%	69 (15.8%)
AST Fallback (Line 50)	0 (0/0)	0.0 $\pm$ 0.0	0.0 (0)	0.0 (0)	100.0%	100.0%	0.0%	0 (0.0%)
AST Combined (Full Branch)	438 (131/162)	26.7 $\pm$ 23.3	250.8 (171)	290.1 (213)	78.1%	87.2%	100.0%	69 (15.8%)

Note: Evaluated over 40 GitHub repositories (189 source files). Tokenizer: `BAAI/bge-m3` (maximum context length 512). Chunk count breakdown: Total (Java / React-TS). Boundary Preservation reflects percentage of chunks not artificially sliced across syntactic statement boundaries. Clean Parse denotes syntactically valid code blocks parsable by tree-sitter without syntax errors. Data dynamically derived from `table_3_2_measured.json` and `chunk_corpus.parquet`.

III. RETRIEVAL EFFECTIVENESS BENCHMARK (2X2 FACTORIAL)

Table III presents the complete 2×2 factorial evaluation. Across the primary Re-ranking tier, Line With-Header achieves NDCG@10 of 0.621 [0.555, 0.682], followed by Line No-Header at 0.580 [0.505, 0.649], AST With-Header at 0.487 [0.419, 0.551], and AST No-Header at 0.341 [0.263, 0.421].

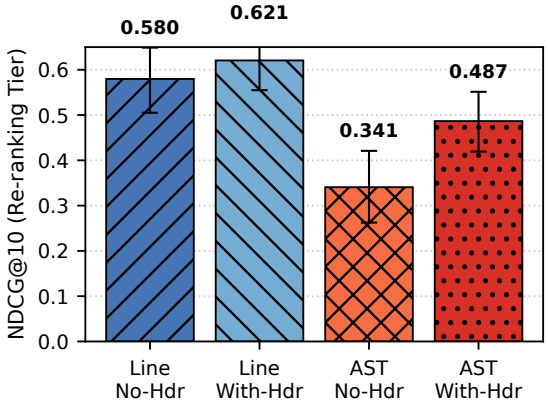


Fig. 2. NDCG@10 retrieval effectiveness across the four conditions at the two-stage re-ranking tier. Error bars represent 95% bootstrap confidence intervals across 25 JDs.

Figure 2 illustrates the re-ranking performance with 95% bootstrap confidence intervals. Figure 3 demonstrates the pronounced interaction effect: while context headers yield a modest non-significant improvement on line chunks ($\Delta = +0.041$), they recover $+0.146$ NDCG@10 on AST chunks ($p_{\text{Holm}} = 0.0001$).

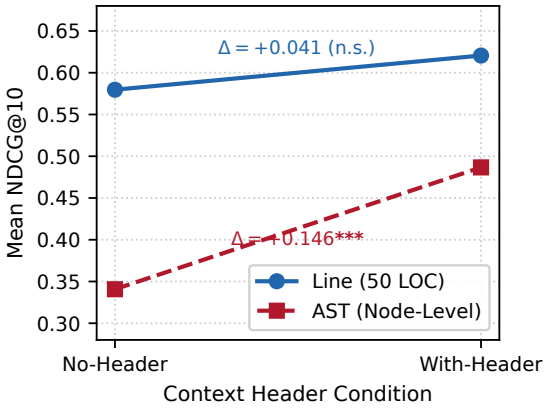


Fig. 3. Interaction plot demonstrating differential context header sensitivity between Line-based chunking and AST-based chunking.

TABLE III

RETRIEVAL EFFECTIVENESS ACROSS CHUNKING STRATEGIES, CONTEXT HEADER CONDITIONS, AND PIPELINE TIERS ($N = 25$ JOB DESCRIPTIONS).

Chunking Strategy	Header Condition	NDCG@10↑	NDCG@1↑	NDCG@3↑	NDCG@5↑	MRR@10↑	P@5↑	Recall@10↑	Judged@10
Panel A: Two-Stage Re-ranking Tier (Cross-Encoder BGE-Reranker-Base)									
Line (50 LOC)	No-Header	0.580 [0.505, 0.649]	**0.800**	0.736	0.728	**0.953**	0.816	0.230	0.716
Line (50 LOC)	With-Header	**0.621** [0.555, 0.682]	**0.800**	**0.773**	**0.770**	0.950	**0.896**	**0.262**	**0.736**
AST (Node-Level)	No-Header	0.341 [0.263, 0.421]	0.340	0.346	0.375	0.699	0.544	0.156	0.712
AST (Node-Level)	With-Header	0.487 [0.419, 0.551]	0.740	0.698	0.661	0.896	0.744	0.183	0.660
Panel B: Dense Bi-Encoder Tier (BGE-M3 Embeddings)									
Line (50 LOC)	No-Header	0.521 [0.439, 0.600]	0.600	**0.612**	0.615	0.891	0.792	0.225	0.720
Line (50 LOC)	With-Header	**0.551** [0.480, 0.619]	0.580	0.609	**0.628**	0.883	**0.832**	**0.265**	**0.752**
AST (Node-Level)	No-Header	0.358 [0.298, 0.420]	0.420	0.381	0.412	0.813	0.608	0.167	0.716
AST (Node-Level)	With-Header	0.450 [0.393, 0.506]	**0.620**	0.589	0.581	**0.910**	0.784	0.204	0.692
Panel C: Lexical Sparse Baseline Tier (BM25)†									
Line (50 LOC)	No-Header	0.405 [0.350, 0.458]	**0.440**	0.502	**0.523**	0.865	**0.824**	0.208	0.644
Line (50 LOC)	With-Header	**0.433** [0.376, 0.486]	**0.440**	**0.506**	0.520	**0.868**	0.800	**0.228**	**0.696**
AST (Node-Level)	No-Header	0.305 [0.236, 0.371]	0.340	0.414	0.429	0.724	0.592	0.128	0.560
AST (Node-Level)	With-Header	0.311 [0.243, 0.376]	0.340	0.309	0.353	0.761	0.576	0.153	0.676
Pre-registered Random Baselines									
Random Selection	Full Corpus (872 Chunks)	0.022	—	—	—	—	—	—	—
Random Selection	Judged Pool Candidates‡	0.562	—	—	—	—	—	—	—

Note: 95% bootstrap confidence intervals across 25 JDs (10,000 resamples, seed 42) shown in brackets below NDCG@10. In each panel, **bold** denotes the best performing configuration, and underline denotes the second-best configuration. Linear relevance gain $\mathrm{rel}/\log_2(i+1)$ is employed across all NDCG calculations.

All 12 configurations across lexical (BM25), dense bi-encoder (BGE-M3), and two-stage cross-encoder (BGE-Reranker-Base) tiers were pooled uniformly at Top-5 ($N = 25$ JDs, $N = 882$ unique graded candidate spans, Judged@5 = 100.0%). All metrics are evaluated against the adjudicated unified ground truth.

‡ Random selection within pool samples strictly from pre-judged items (rel $\in \{0, 1, 2\}$) and is non-comparable to open-retrieval systems facing unjudged candidates.

IV. HYPOTHESIS TESTING AND PAIRED COMPARISONS

Table IV presents the non-parametric Wilcoxon signed-rank tests for the confirmatory family and the exploratory contrast.

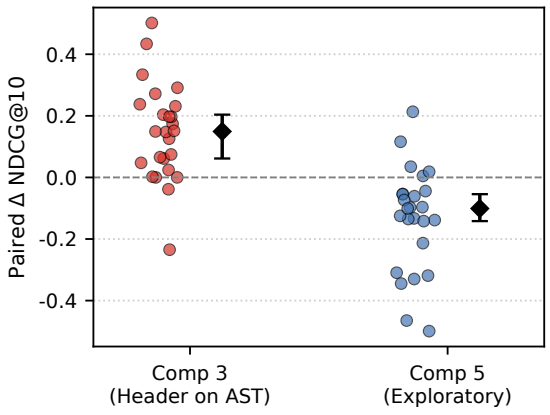


Fig. 4. Per-query paired difference (Δ NDCG@10) strip plot for Comp 3 (AST Header Effect) and Comp 5 (Exploratory AST vs. Line+Header). Black diamonds denote median and 95% bootstrap CI.

- **Comp 1 (Full Proposed vs. Baseline):** The overall difference between AST+Header and Line No-Header is $\Delta = -0.0930$ (95% CI [-0.164, -0.023], $p_{\text{raw}} = 0.0219$, $p_{\text{Holm}} = 0.0438$). The pre-registered primary comparison is statistically significant, favoring Line No-Header over AST+Header.
- **Comp 2 (Pure Chunking):** In the absence of context headers, AST chunking underperforms Line chunking by $\Delta = -0.2390$ ($p_{\text{Holm}} < 0.001$).
- **Comp 3 (Header on AST):** Adding structural context headers to AST chunks produces a large, statistically significant gain of $\Delta = +0.1459$ ($p_{\text{Holm}} = 0.0001$).
- **Comp 4 (Header on Line):** Context headers produce a minor, non-significant change on Line chunks ($\Delta = +0.0409$, $p_{\text{Holm}} = 0.2411$).
- **Comp 5 (Exploratory Contrast):** Line With-Header achieves an edge of 0.1339 over AST With-Header ($p_{\text{raw}} = 0.0003$, 95% CI [-0.199, -0.070]), which is statistically significant.

Figure 4 displays the per-query paired differences for Comp 3 (23 wins, 2 losses) and Comp 5 (5 wins, 20 losses, 0 ties).

TABLE IV
HYPOTHESIS TESTING RESULTS FOR PRE-REGISTERED PRIMARY COMPARISONS AND EXPLORATORY CONTRAST (NDCG@10, RE-RANKING TIER).

ID	Hypothesis Description	Contrast ($A - B$)	Δ NDCG@10	95% Bootstrap CI	W/L/T	W	p_{raw}	p_{Holm}
Pre-registered Confirmatory Family (Step-down Holm-Bonferroni Correction, $\alpha = 0.05$)								
Comp 1	Full Proposed vs. Full Baseline	AST+Header vs. Line No-Header	-0.0930	[-0.164, -0.023]	7/18/0	78.0	0.0219	0.0438^{*}
Comp 2	Pure Chunking Effect	AST No-Header vs. Line No-Header	-0.2390	[-0.292, -0.185]	2/23/0	3.0	< 0.001	$< 0.001^{***}$
Comp 3	Context Header Effect on AST	AST+Header vs. AST No-Header	$+0.1459$	[0.086, 0.207]	23/2/0	24.0	< 0.001	$< 0.001^{***}$
Comp 4	Context Header Effect on Line	Line+Header vs. Line No-Header	$+0.0409$	[-0.012, 0.093]	15/10/0	118.0	0.2411	0.2411
Exploratory Analysis (Outside Pre-registered Family)								
Comp 5	AST+Header vs. Line+Header	AST+Header vs. Line+Header	-0.1339	[-0.199, -0.070]	5/20/0	36.0	< 0.001	—

Note: $N = 25$ paired Job Descriptions. W: Wilcoxon signed-rank test statistic. Win/Loss/Tie (W/L/T) indicates the count of queries where System $A > B$, $A < B$, or $A = B$. Holm adjusted p-values (p_{Holm}) control family-wise error rate across Comps 1–4. Significance levels: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Comp 5 is an exploratory comparison outside the frozen registered confirmatory family and is not adjusted under Holm's step-down procedure.

V. SUBGROUP STRATIFICATION AND SENSITIVITY ANALYSES

Table V stratifies performance across Java ($N = 15$) and React/TypeScript ($N = 10$). In Java, AST+Header reaches 0.5830 vs. 0.7158 for Line+Header ($\Delta = -0.133$). In React/TypeScript, AST+Header reaches 0.3420 vs. 0.4777 for Line+Header ($\Delta = -0.136$).

Table VI tracks the composition of retrieved top-10 items. Following the AST parser upgrade, 0 (0.0%) fallback chunks appear in the top-10 across all tiers.

Table VII presents sensitivity audits. When removing all 17 disputed items (Panel A: Consensus Only, $N = 865$), the AST header effect remains large and significant ($\Delta = +0.1528$, $p < 0.001$). Under judged-only evaluation (Panel C), Line+Header achieves 0.7243 vs. 0.5774 for AST+Header.

TABLE V
RETRIEVAL PERFORMANCE (NDCG@10, RE-RANKING TIER) STRATIFIED BY TARGET PROGRAMMING LANGUAGE.

Configuration	Java ($N = 15$)	React/TS ($N = 10$)	Overall ($N = 25$)
Line No-Header	<u>0.695</u>	<u>0.407</u>	<u>0.580</u>
Line With-Header	**0.716**	**0.478**	**0.621**
AST No-Header	0.470	0.147	0.341
AST With-Header	0.583	0.342	0.487
Δ (AST+Hdr vs. Line+Hdr)†	-0.1328	-0.1357	-0.1339

Note: Stratification over 15 Java job descriptions and 10 React/TypeScript job descriptions at the two-stage re-ranking tier. **Bold** denotes the best performing configuration, and <u>underline</u> denotes the second-best within each language stratum. NDCG computed using linear gain $\mathrm{rel}/\log_2(i+1)$.
† Exploratory contrast (Comp 5). Negative values denote advantage for Line With-Header.

TABLE VI
AST FALLBACK COMPOSITION IN TOP-10 RETRIEVAL RESULTS ACROSS PIPELINE TIERS AND HEADER CONDITIONS.

Pipeline Tier	Header Condition	Total Top-10 (N)	AST Pure (%)	Line-Fallback (%)
BM25	No-Header	250	250 (100.0%)	0 (0.0%)
BM25	With-Header	250	250 (100.0%)	0 (0.0%)
Dense (BGE-M3)	No-Header	250	250 (100.0%)	0 (0.0%)
Dense (BGE-M3)	With-Header	250	250 (100.0%)	0 (0.0%)
Re-ranking (Cross-Encoder)	No-Header	250	250 (100.0%)	0 (0.0%)
Re-ranking (Cross-Encoder)	With-Header	250	250 (100.0%)	0 (0.0%)

Note: Evaluated across 25 JDs (250 total top-10 slots per configuration). Line-Fallback chunks occur when AST parsing fails or exceeds maximum LOC thresholds, defaulting to sliding-window chunks. Derived dynamically from `sensitivity_analysis_report.json`.

TABLE VII
SENSITIVITY ANALYSIS ACROSS GROUND TRUTH CURATION SUBSETS, SINGLE-ANNOTATOR QRELS, AND JUDGED-ONLY POOLS.

Evaluation Subset	Line+Hdr	Line No-Hdr	AST+Hdr	AST No-Hdr	AST Header Gain (Δ)	Statistical Decision
Panel A: Consensus-Only Ground Truth (17 Disagreements Excluded, $N = 865$ Pairs)						
Consensus Pairs	0.617	0.571	0.487	0.334	$+0.1528$ ($p < 0.001$)	Significant ($p < 0.001$)
Panel B: Single-Annotator Ground Truth Variants						
Annotator 1 Only	—	—	0.486	0.341	$+0.1446$ ($p < 0.001$)	Significant ($p < 0.001$)
Annotator 2 Only	—	—	0.486	0.338	$+0.1477$ ($p < 0.001$)	Significant ($p < 0.01$)
Panel C: Judged-Only Candidate Evaluation (Unjudged Candidates Removed)						
Judged-Only Pool	0.724	0.670	0.577	0.416	$+0.1618$ ($p < 0.001$)	Significant ($p < 0.01$)

Note: NDCG@10 at the two-stage re-ranking tier computed using linear gain $\mathrm{rel}/\log_2(i+1)$. In Panel A, removing disputed items yields an AST header effect of $\Delta = +0.1528$ ($p < 0.001$). Data derived dynamically from `sensitivity_analysis_report.json`.

VI. CANDIDATE POOL AUDIT AND DATASET PROVENANCE

Table VIII provides a complete coverage audit. Top-3 and Top-5 judgment coverage was 100.0% across all 12 configurations.

Table IX details pool deduplication (Panel A) and adjudication decisions (Panel B). 1500 raw Top-5 instances collapsed into 882 unique candidate spans. Adjudication resolved 17 disagreements into 13 agreements with Annotator 1, 4 with Annotator 2, and 0 independent resolutions.

Table X summarizes open-source licenses across the 40 mined evaluation repositories (19 MIT, 16 unspecified public GitHub, 4 Apache-2.0, 1 AGPL-3.0).

TABLE VIII
CANDIDATE POOL JUDGMENT COVERAGE AND UNJUDGED CANDIDATE EXPOSURE ACROSS 12 RETRIEVAL CONFIGURATIONS.

Configuration ID	Pipeline Tier	Top-3 Evaluated	Top-3 Unjudged	Judged@3 (%)	Top-10 Evaluated	Top-10 Unjudged	Judged@10 (%)
bm25_line_no_header	BM25†	75	0	100.0%	250	89	64.4%
bm25_line_with_header	BM25†	75	0	100.0%	250	76	69.6%
bm25_ast_no_header	BM25†	75	0	100.0%	250	110	56.0%
bm25_ast_with_header	BM25†	75	0	100.0%	250	81	67.6%
dense_line_no_header	Dense	75	0	100.0%	250	70	72.0%
dense_line_with_header	Dense	75	0	100.0%	250	62	75.2%
dense_ast_no_header	Dense	75	0	100.0%	250	71	71.6%
dense_ast_with_header	Dense	75	0	100.0%	250	77	69.2%
rerank_line_no_header	Rerank	75	0	100.0%	250	71	71.6%
rerank_line_with_header	Rerank	75	0	100.0%	250	66	73.6%
rerank_ast_no_header	Rerank	75	0	100.0%	250	72	71.2%
rerank_ast_with_header	Rerank	75	0	100.0%	250	85	66.0%

Note: Evaluated across 25 JDs (75 total top-3 slots, 125 total top-5 slots, 250 total top-10 slots). Pool judgment coverage was 100.0% at Judged@3 and Judged@5 across all 12 retrieval configurations (BM25, Dense, and Re-ranking). Derived dynamically from `pool_coverage_report.csv`.

TABLE IX
POOL DEDUPLICATION RECONCILIATION AND ADJUDICATION DECISION DISTRIBUTION.

Reconciliation / Adjudication Stage	Count	Proportion (%)
Panel A: Coordinate Deduplication & Pool Construction (Top-5 across 12 Systems)		
Raw Pooled Candidate Instances ($25 \times 12 \times 5$)	1500	—
Unique Deduplicated Candidate Spans	882	58.8%
Multi-Chunk Collapsed Coordinates	1	0.1%
Complete Unified Ground Truth ($N = 882$ pairs)	882	100.0%
Panel B: Discrepancy Adjudication Decisions ($N = 17$ Disagreements)		
Agreed with Annotator 1	13	76.5%
Agreed with Annotator 2	4	23.5%
Both Overruled (Independent Resolution)	0	0.0%
Structured Resolution Reasons (Distinct Strings)	15	—
— Explicit Prefix `RESOLVED_BY_ADJUDICATION`	17	100.0%
— Standard Evaluation Record Format	0	0.0%

Note: In Panel A, 1500 raw Top-5 retrieval candidates across 12 configurations collapsed into 882 unique code spans. In Panel B, adjudication was performed by an independent lead adjudicator following structured templates. Derived from `pool_reconciliation_report.txt` and `adjudication_summary.json`.

TABLE X
DISTRIBUTION OF OPEN-SOURCE SOFTWARE LICENSES ACROSS THE 40 MINED EVALUATION REPOSITORIES.

License Category	Repositories (n)	Proportion (%)	Legal Classification
MIT License	19	47.5%	Permissive
Public GitHub (Not specified)	16	40.0%	Default Copyright
Apache-2.0	4	10.0%	Permissive
GNU AGPL-3.0	1	2.5%	Strong Copyleft
Total Evaluated Repositories	40	100.0%	20 Java / 20 React

Note: Tabulated from `manifest_repos.csv`. Mined snippets are distributed solely for peer-reviewed academic replication under scientific fair use.

APPENDIX A
CRYPTOGRAPHIC FILE PROVENANCE

Table XI records the SHA-256 digests and timestamps of all 14 data and result artifacts utilized to generate this report.

TABLE XI
CRYPTOGRAPHIC VERIFICATION AND TIMESTAMPS OF SOURCE FILES.

Relative File Path	SHA-256 (12)	Last Modified
dataset/ground_truth_annotator1.csv	54f76fe2219b	2026-10-21 22:11
dataset/ground_truth_annotator2.csv	6cb859250fb3	2026-10-21 22:11
dataset/ground_truth_final.csv	0dea15e2b45b	2026-10-21 22:11
dataset/chunk_corpus.parquet	918cd8263d92	2026-10-21 22:11
dataset/table_3_2_measured.json	ec60df03ab0c	2026-10-21 22:11
dataset/table_3_2_report.txt	2595e43b5ee9	2026-10-21 22:11
dataset/benchmark_results/table_3_retrieval_benchmark.csv	077875b2b030	2026-10-21 22:11
dataset/benchmark_results/per_query_evaluation.csv	e3c86b69b59b	2026-10-21 22:11
dataset/benchmark_results/statistical_tests_report.json	8eda00a0e496	2026-10-21 22:11
dataset/benchmark_results/sensitivity_analysis_report.json	894c4e7aa274	2026-10-21 22:11
dataset/benchmark_results/pool_coverage_report.csv	42b9531ea8f2	2026-10-21 22:11
dataset/benchmark_results/pool_reconciliation_report.txt	97a2e8568966	2026-10-21 22:11
dataset/benchmark_results/adjudication_summary.json	d22068548919	2026-10-21 22:11
dataset/manifest_repos.csv	7bd2b90ca5a5	2026-10-21 22:11

APPENDIX B
REPRODUCTION INSTRUCTIONS

To reproduce all numbers, tables, figures, and compile this report from scratch, execute:

```
# 1. Run evaluation & tests
python scripts/evaluate_retrieval.py
python scripts/evaluate_sensitivity.py

# 2. Build report artifacts
python scripts/make_report.py

# 3. Verify numerical concordance
python -m pytest tests/test_report_numbers.py
```